

Mike Roe Mouse: Parts, Specifications, and Schematics

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1 Overview

This micromouse is being developed for the 2027 APEC MicroMouse competition in March 2027.

1.1 Software Tools Overview

The following software tools are being used.

CAD	OnShape
PCB/Schematic	KiCAD
Repository	Github
IDE	STM32CubeIDE

1.2 Parts Overview

The following parts will be used.

Microcontroller	STM32F405RGT6
Oscillator	TBD
Bluetooth Microcontroller	nRF52810
Buck Converter	TPS564242
MOSFET	IRLR024NTRPBF
IR Emitter	SFH4545
IR Phototransistor	TEFT4300
Red Emitter	Check
Red Phototransistor	TBD
Battery	URGENEX 7.4V 2S 60C
IMU	ICM-42670-P

2 Microcontroller Parts

2.1 STM32F405RGT6 Microcontroller

The F-series STM32 microcontroller will handle all computational tasks. This includes navigation and maze-solving algorithms, IR emitter control, and vacuum fan control.

2.1.1 Purchasing Information

As of July 9th, 2026

Name	Per Unit	Quantity	Total	Vendor	Link
STM32F405RGT6	\$5.85	1	\$5.85	JLC	Link

2.1.2 Dimensions

10 mm x 10 mm

2.1.3 Electrical Specifications

I/O pins CMOS and TTL compliant. Not all I/O pins are 5V compliant.
Logic level voltage values from p.110 of data sheet.

Logic Level	Logic Type	Min	Max
V_{IH}	CMOS	$0.7V_{DD}$	5.0V
V_{IL}	TTL	0V	0.8V
V_{IH}	TTL	2.0V	5.0V

Maximum voltage and current ratings for V_{DD} and I/O.
 V_{DD} ranges are with respect to ground indicated by V_{SS} .
 V_{DD} ranges and I/O ranges on pp. 76–77 of data sheet.

Max V_{DD}	4.0V
Min V_{DD}	-0.3V
Max $I_{V_{DD}}$ Sink	150mA
Max $I_{V_{SS}}$ Source	150mA
Max $I_{I/O}$ Sink	25mA
Max $I_{I/O}$ Source	25mA

2.2 INSERT OSCILLATOR

2.3 8520 Fan Motor

3 Arena Sensing Parts

3.1 SFH4545 Infrared Emitter

Infrared emitter that the micromouse uses to detect distance from walls.
Distance calculated based on return time for infrared emitter.

3.1.1 Purchasing Information

As of July 17th, 2026

Name	Price Per Unit	Quantity	Total Cost	Vendor	Item Link
SFH4545	\$0.79	4	\$3.16	DigiKey	Link

3.1.2 Electrical Specifications

Operating current and voltage requirements. On pp. 4–5 of spec sheet.
All voltages and currents listed assume a forward direction through the intended direction through the device.

Max $I_{Constant}$	100mA
Max I_{Pulsed}	1A
Max $V_{Constant}$	1.7V
Max V_{Pulsed}	2.9V

Operational centroid wavelengths

Minimum	Typical	Maximum
920nm	940nm	960nm

4 Power Electronics and Battery

4.1 TPS564242 Buck Converter

4.2 IRLR024NTRPBF N-Channel MOSFET

4.3 Insert Schottky Diode

5 Drivetrain